Knapsack(0/1) - Dynamic programming

Name: Mistry Nishk Nileshkumar
Registration Number: 24BCE5236
Course: Design Analysis and Algorithm
Code: BCSE204P
Faculty: Manimegalai

Code:

```
#include<bits/stdc++.h>
using namespace std;
int main(){
    int n;
    long long W;
    cout << "Name: Mistry Nishk Nileshkumar\nRegistration Number: 24BCE5236\n";
    cout << "Enter number of items: ";
    cin >> n;
    cout << "Enter capacity: ";
    cin >> W;
    struct Item{ long long w; long long v; double r; };
    vector<Item> items;
    long long zeroValueSum=0;
    cout << "Enter value and weight: (v w)\n";
    for(int i=0;i<n;i++){
        long long w,v;
        cin>>v>>w;
        if(w==0){
            zeroValueSum += v;
        } else {
            items.push_back({w,v, (double)v / (double)w});
        }
    }
    sort(items.begin(), items.end(), [](const Item &a, const Item &b){
        if(a.r!=b.r) return a.r>b.r;
        if(a.v!=b.v) return a.v>b.v;
        return a.w<b.w;
    });
    double rem = (double)W;
    double fracTotal = (double)zeroValueSum;
    for(auto &it: items){
        if(rem<=0) break;
        double take = min(rem, (double)it.w);
        fracTotal += take * it.r;
        rem -= take;
    }
    vector<long long> dp(W+1,0);
    for(auto &it: items){
        long long wt = it.w;
        long long val = (long long) it.r * 0 + (long long) llround(it.r * 0) ;
        val = (long long) (it.r * it.w + 0.5);
        val = (long long) (it.r * it.w);
```

```
        val = (long long) (it.r * it.w);
        val = (long long) (it.r * it.w);
        val = (long long) (it.r * it.w);
        val = (long long) (it.r * it.w);
        val = (long long) (it.r * it.w);
        val = (long long) (it.r * it.w);
        val = (long long) llround(it.r * it.w);
        val = (long long) (it.r * it.w);
        val = (long long) (it.r * it.w);
        val = (long long) (it.r * it.w);
        val = (long long) (it.r * it.w);
        val = (long long) (it.r * it.w);
        val = (long long) (it.r * it.w);
        val = (long long) (it.r * it.w);
        val = (long long) llround(it.r * it.w);
        for(long long cap=W; cap>=wt; --cap){
            dp[cap] = max(dp[cap], dp[cap-wt] + val);
        }
    }
    long long bestDP = dp[W] + zeroValueSum;
    cout << "Maximum Value: " << bestDP;
    return 0;
}
```

Input & Output:

```
C:\Windows\System32\cmd.e

C:\Users\Nishk\Documents\DAA>g++ greedy_knapsack_01.cpp -o greedy_knapsack_01

C:\Users\Nishk\Documents\DAA>greedy_knapsack_01.exe
Name: Mistry Nishk Nileshkumar
Registration Number: 24BCE5236
Enter number of items: 4
Enter capacity: 10
Enter value and weight: (v w)
3 2
4 3
5 4
8 6
Maximum Value: 13
C:\Users\Nishk\Documents\DAA>
```